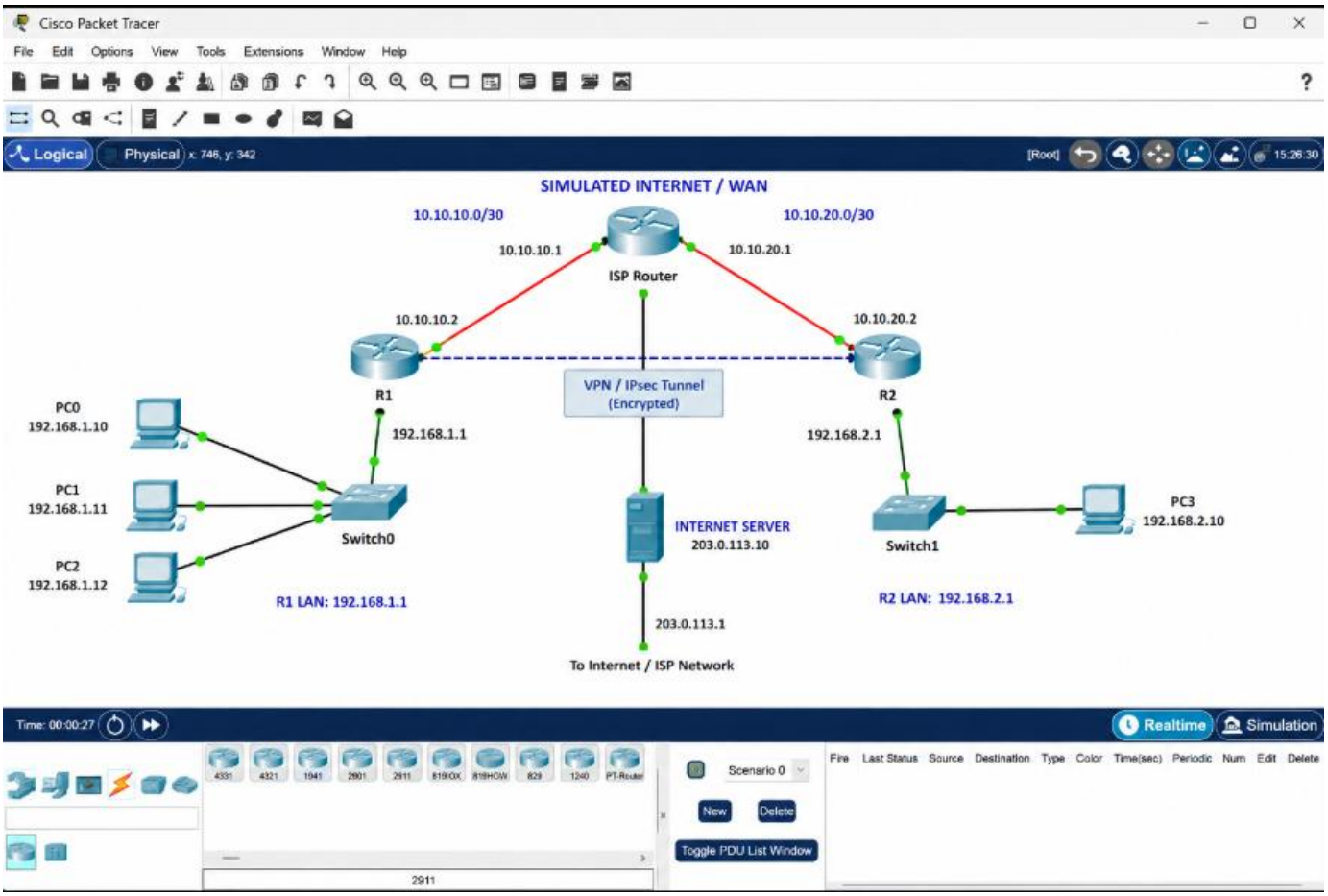
DAY_4

- **Weekly Practice and Troubleshooting Practical: Simulate a WAN setup with VPN and NAT. Task: Troubleshoot connectivity issues in the setup Checkpoint: Quiz: WAN technologies, VPNs, and NAT/PAT Practical Evaluation: Configure and troubleshoot a WAN.**

1. Use Cisco Packet Tracer to simulate a WAN setup connecting two branch offices via a VPN tunnel and configure NAT on the edge routers so both branches can access the internet.

ANS :

- Create a simulated WAN connecting **two branch offices** through a VPN tunnel. Configure **PAT/NAT** so internal devices can access an external Internet server. Then intentionally create VPN and routing problems and troubleshoot them using show and debug commands.



- **R1 Configuration**

```
enable
configure terminal

hostname R1

interface gigabitEthernet0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet0/1
ip address 10.10.10.2 255.255.255.252
no shutdown
exit

    ip route 0.0.0.0 0.0.0.0 10.10.10.1
```

- **R2 Configuration**

```
enable
configure terminal

hostname R2

interface gigabitEthernet0/0
ip address 192.168.2.1 255.255.255.0
no shutdown
exit

interface gigabitEthernet0/1
ip address 10.10.20.2 255.255.255.252
no shutdown
exit

    ip route 0.0.0.0 0.0.0.0 10.10.20.1
```

2. Configure Port Address Translation (PAT) on a router in your simulated WAN so that three devices from the internal network can browse the internet using a single public IP address. Hint: Use the 'overload' keyword in your NAT configuration.

ANS :

Device	IP Address	Gateway
PC0	192.168.1.10	192.168.1.1
PC1	192.168.1.11	192.168.1.1
PC2	192.168.1.12	192.168.1.1

- **Configure NAT/PAT on R1**

```
enable
configure terminal

interface gigabitEthernet0/0
ip nat inside
exit

interface gigabitEthernet0/1
ip nat outside
exit

access-list 1 permit 192.168.1.0 0.0.0.255

ip nat inside source list 1 interface gigabitEthernet0/1 overload

end
    write memory
```

- **PAT Result**

```
PC0 192.168.1.10 ─┐
PC1 192.168.1.11 ─┼── PAT Overload ──→ 10.10.10.2
PC2 192.168.1.12 ─┘
```

3. Intentionally misconfigure the VPN tunnel's pre-shared key between two routers in your WAN simulation, then use the 'show crypto isakmp sa' and 'show run' commands to identify and fix the connectivity issue.

ANS :

- **Configure GRE + IPsec VPN**
  - Step 1: Create the GRE Tunnel
  - Step 2: Configure ISAKMP Phase 1
  - Step 3: Configure IPsec

- **Router1**

```
enable
configure terminal

! ================================
! STEP 1: GRE Tunnel Configuration
! ================================

interface tunnel0
ip address 172.16.10.1 255.255.255.252
tunnel source gigabitEthernet0/1
tunnel destination 10.10.20.2
no shutdown
exit

! Route to Branch 2 through GRE Tunnel
ip route 192.168.2.0 255.255.255.0 172.16.10.2

! ================================
! STEP 2: ISAKMP Phase 1
! ================================

crypto isakmp policy 10
encryption aes
hash sha
authentication pre-share
group 2
exit

! Pre-shared Key
crypto isakmp key VPNKEY123 address 10.10.20.2

! ================================
```

```
! STEP 3: IPsec Configuration
! ================================

crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
mode transport
exit

crypto ipsec profile VPN-PROFILE
set transform-set VPN-SET
exit

! Apply IPsec Protection to GRE Tunnel
interface tunnel0
tunnel protection ipsec profile VPN-PROFILE
exit

end
    write memory
```

- **Router2**

```
enable
configure terminal

! ================================
! STEP 1: GRE Tunnel Configuration
! ================================

interface tunnel0
ip address 172.16.10.2 255.255.255.252
tunnel source gigabitEthernet0/1
tunnel destination 10.10.10.2
no shutdown
exit

! Route to Branch 1 through GRE Tunnel
ip route 192.168.1.0 255.255.255.0 172.16.10.1

! ================================
! STEP 2: ISAKMP Phase 1
! ================================

crypto isakmp policy 10
encryption aes
hash sha
authentication pre-share
group 2
exit
```

```
! Pre-shared Key
crypto isakmp key VPNKEY123 address 10.10.10.2

! ================================
! STEP 3: IPsec Configuration
! ================================

crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
mode transport
exit

crypto ipsec profile VPN-PROFILE
set transform-set VPN-SET
exit

! Apply IPsec Protection to GRE Tunnel
interface tunnel0
tunnel protection ipsec profile VPN-PROFILE
exit

end
    write memory
```

4. Troubleshoot a scenario where devices in one branch office can ping the internet but cannot access devices in the other branch via the VPN tunnel; document the commands you use and the configuration change that resolves the problem.

ANS :

- **Problem Scenario**
  - Suppose PCs in Branch 1 can ping the Internet server:
    PC0 → 203.0.113.10 = SUCCESS
  - But cannot ping Branch 2:
    PC0 → 192.168.2.10 = FAILED
- **Troubleshooting Procedure**
  - **Step 1: Check Interfaces**

    show ip interface brief
    Verify:
    - LAN interface is up/up
    - WAN interface is up/up
    - Tunnel0 is up/up

- **Step 2: Check Tunnel Status**
  show interface tunnel0
  Verify the tunnel source and destination are correct.

- **Step 3: Check VPN Status**
  show crypto isakmp sa
  show crypto ipsec sa
  If the security association is not established, investigate the ISAKMP/IPsec configuration.

- **Step 4: Check Routing**

  - On R1:
    show ip route

  - The routing table must contain a route for:
    192.168.2.0/24

  - The correct route is:
    S 192.168.2.0/24 [1/0] via 172.16.10.2

  - If the route is missing, add:
    configure terminal
    ip route 192.168.2.0 255.255.255.0 172.16.10.2
    end

  - On R2, verify the return route:
    show ip route
    The correct route should be:
    S 192.168.1.0/24 [1/0] via 172.16.10.1

  - If missing:
    configure terminal
    ip route 192.168.1.0 255.255.255.0 172.16.10.1
    end

➢ **Main Configuration Change That Resolves the Issue**

  - The problem can occur when the branch-to-branch static routes are missing.
  - **Add on R1**
    ip route 192.168.2.0 255.255.255.0 172.16.10.2
    Add on R2
    ip route 192.168.1.0 255.255.255.0 172.16.10.1

  - After adding both routes:
    Branch 1 PC → R1 → GRE/IPsec Tunnel → R2 → Branch 2 PC
    Connectivity should work.

5. Take the WAN, VPN, and NAT scenario you built and swap your configuration file with a classmate; review their setup, find one misconfiguration, and write a short note on how you identified and fixed it. Constraint: Focus on using 'show' and 'debug' commands for troubleshooting rather than guessing the issue.

ANS :

- **Example Misconfiguration Found**

  show running-config
  show ip route
  show ip interface brief
  show crypto isakmp sa
  show crypto ipsec sa

  - The missing route was:
    ip route 192.168.1.0 255.255.255.0 172.16.10.1
  - I added the route and tested the connection again using:
    ping 192.168.1.10
  - After adding the correct return route, the branch-to-branch connectivity was restored.

- **Complete Troubleshooting Commands**

  show ip interface brief
  show interfaces tunnel0
  show ip route
  show running-config
  show running-config | section crypto
  show running-config | include crypto isakmp key
  show crypto isakmp sa
  show crypto ipsec sa
  show ip nat translations
  show ip nat statistics
  show access-lists